

CSA0911
Java

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COI-ATB-3

1) Student Management System

class Student {

private String name;

private int rollNo;

private int marks;

Student(String name, int rollNo, int marks) {

this.name = name;

this.rollNo = rollNo;

this.marks = marks;

void markAvg(marks) {

int Avg = marks / 6

System.out.println("Avg is " + Avg)

if (marks > 60)

S.O. println("PASS")

else

S.O.P("Fail")

// End class

Class Main

public static void main(String[] args)

Student S1 = new Student("Uamni", 1234, 700)

Student S2 = new Student("Pradeep", 1235, 80)

Call markAvg()

End class

2) Library Management System

class Library

private String title

private String Author

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```
private int ISBN
private boolean isAvail
```

```
Library(String title, String Author, int ISBN, boolean isAvail)
{
    this.title = title;
    this.Author = Author;
    this.ISBN = ISBN;
    this.isAvail = true;
}
```

```
void bookIssue()
```

```
{
    if (isAvail)
```

```
    {
        print "Book issued"
```

```
    }
    else
```

```
    {
        print "Not available"
```

```
    }
}
```

```
if (!isAvail)
```

```
    {
        print "Book is returned"
```

```
    }
    else
```

```
    {
        print "Book has not returned"
```

```
End class
```

```
class Main
```

```
{
    public Main method()
```

```
    {
        Library L = new L("Udumra", "Udumra", 108, true);
```

```
        L.bookIssue();
```

```
        L.returnBook();
```

```
        // print Book details
```

```
        L.getTitle();
```

```
        L.getAuthor();
```

```
        L.getISBN();
```

```
    }
}
```

3) Employee salary calculation

```
class Salary
```

```
{
    private int empId
```

```
private String name
```

```
private int empId
```

```
private String designation
```

```
public String getName()
```

```
{
    return name
```

```
}
public String getDesignation()
```

```
{
    return designation
```

```
}
void GrossSalary(int basic, int HRA, int DA)
```

```
{
    GTS = basic + HRA + DA
```

```
    print GTS
```

```
void netSalary(int GTS, PF, tax)
```

```
{
    net = GTS - PF - tax
```

```
    print net
```

```
void display()
```

```
{
    print name
```

```
    print basic pay
```

```
    print designation
```

```
main class
main method()
```

```
{
    Salary S = new Salary()
```

```
    {
        S.setName("Suresh")
```

```
        S.setBasicPay(75000)
```

```
        S.setDesignation("Software Engineer")
```

```
        S.display()
```

```
        S.netSalary()
```

```
        S.grossSalary()
```

```
    }
}
```

```
End class
```

1) Shape Area calculation using polymorphism

class Shape

public int radius

public int length

public int breadth

public int area

public int height

end class

class Circle extends Shape

function ^{def} calculateArea()

return $\pi \times \text{radius} \times \text{radius}$

end function

end class

class Rectangle extends Shape

public int length

public int

breadth

end function

public int calculateArea()

return $\text{length} \times \text{breadth}$

end function

end class

main class

main method

Shape s1 = new Circle(5)

Shape s2 = new Rectangle(5, 10)

Shape s3 = new Shape(5, 10)

2) Vehicle Registration System

class Vehicle

public int engine

public String color

public String fuelType

void displayDetails()

get "details of vehicle is"

class TwoWheeler extends Vehicle

private int engineCapacity

private int seatCap

private int noOfWheels

void displayDetails()

get "capacity is" engineCapacity

get "type of vehicle" seatCap

get "number of wheels" noOfWheels

class FourWheeler extends Vehicle

private int seatCap

void displayDetails()

get "no of wheels is" noOfWheels

end class

main class

main method

Vehicle v1 = new TwoWheeler(10)

Vehicle v2 = new FourWheeler(10)

v1.displayDetails()

v2.displayDetails()

class BankAccount

public accNo

public int balance

public String holder

void withdrawalBehavior()

S.O.P "holder name is" holder

class SavingsAccount extends BankAccount

private int IntPerc

void withdrawalBehavior()

int updated = balance - IntPerc

S.O.P "updated balance is" updated

End class

class CurrentAccount extends BankAccount

private ODL = 5000

public method currentAccount(accNo, bal, holder)

call super (accNo, bal, holder)

End method

void withdrawalBehavior()

if limit > withdraw

S.O.P "Not Eligible"

else

S.O.P eligible

End class

main class

main method

BankAccount B₁ = new SavingsAccount()

BankAccount B₂ = new CurrentAccount()

B₁.withdrawalBehavior(6000)

End class
End method

End

7) Hospital Management System

class Patient

protected PatName

protected patId

private diagnosis

private prescription

public Patient(PatName, PatId, diag, Pres)

this.patName = patName

this.patId = PatId

this.diag = diag

this.pres = pres

End class

class InPatient extends Patient

private wN

private adDate

private charges

public InPatient(wN, adDate, charges)

call super (Name, Id, diag, pres)

wN = wN

adDate = ad

charges = ch

public generateBill()

total = roomBill + doctorFee

print total

End method

class OutPatient extends Patient

private appointmentDate

private Fee

public generateBill()

totalBill = Fee

print totalBill

End method

End class


```
public method calculateA()  
    return nof PHD * 0.05
```

class main

do main method()

Staff S₁ = new TeachingStaff (516, "Anusha", 5000)

Staff S₂ = new Professor (109, "Kumar", 1,00,000)

S₁.calculateAllowance()

S₂.calculateA();

10) class Staff Device

private StaffId

public calculateA()

return D

End class

class Smartlight extends Device

public method execute()

if CMD == "ON" then status "ON"

print name,

End class

class Smartthermostat extends Smartdevice

public method execute command

if Cmd = "ON" then status "ON"

print "AC", status

End class

Main class

main method

d1 = new Smartlight("Room-light")

d2 = new Smarttherm("AC")

d3 = new Smartlock("Door")

d1.executeCommand("ON")

d2.executeCommand("OFF")

End class